

Reinforcement Learning

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The multi-armed bandit (MAB) problem

- ▶ At time t :
- ▶ Select action $a_t \in A$
- ▶ Obtain reward $r_t \in \mathbb{R}$

Basic objective

Maximise total reward

$$U = \sum_{t=1}^T r_t,$$

where T is the **horizon**. It may be unknown, or random.

Regret

We can instead minimise total regret

$$L = \sum_{t=1}^T [r_t^* - r_t],$$

where r^* is the reward an oracle that knew the "best" arm would have obtained.

No let's make this more precise.

The stochastic MAB

For each arm $i \in A$:

- ▶ $r_t \mid a_t = i \sim \mu_i$ is the reward distribution
- ▶ $\rho_i \triangleq E_\mu[r_t \mid a_t = i]$ the expected reward
- ▶ $\rho^* \triangleq \max_i \rho_i$.

Policy

The policy $\pi \in \Pi$ is a adaptive: $\pi(a_t \mid a_{t-1}, r_{t-1}, \dots, a_1, r_1)$

Objective

Maximise expected total reward

$$\mathbb{E}_\mu^\pi[U] = \mathbb{E}_\mu^\pi \left[\sum_{t=1}^T r_t \right]$$

The total expected regret is

$$\mathbb{E}_\mu^\pi[L] = \mathbb{E}_\mu^\pi \left[\sum_{t=1}^T \rho^* - \rho_t \right]$$

The horizon and regret

Discounted T

- ▶ $U = \sum_{t=1}^T \gamma^{t-1} r_t$
- ▶ Same as non-discounted with stopping probability $(1 - \gamma)$.

Arbitrary T

To compare algorithms, we use the notion of regret growth

- ▶ Linear regret: $L_T = O(T)$. i.e. insufficient learning
- ▶ Sub-linear regret, e.g. $L_T = O(\sqrt{T})$ or $O(\ln T)$.

Algorithms

ϵ -greedy

- ▶ $\hat{\rho}_{i,t}$ is the average reward of arm i at time t .
- ▶ w.p. ϵ , $a_t \sim \text{Unif}(A)$
- ▶ otherwise, $a_t = \arg \max_i \hat{\rho}_{i,t}$,

UCB

- ▶ Play all arms once, and for $t > |A|$:
- ▶ $a_t = \arg \max_i \hat{\rho}_{i,t} + \sqrt{2 \ln(t) / n_{i,t}}$.
- ▶ $n_{i,t}$ is the number of times arm i has been pulled until time t .

Thompson (posterior) sampling

Input: a prior β_1 over $\mathcal{M}$.

- ▶ At time t :
- ▶ Sample from the posterior $\mu^{(t)} \mid a_1, r_t, \dots, a_{t-1}, r_{t-1} \sim \beta_t(\mu)$
- ▶ Choose best action for sample: $a_t = \arg \max_i \mathbb{E}_{\mu^{(t)}}[r_t \mid a_t = i]$.
- ▶ Observe r_t .
- ▶ Calculate new posterior $\beta_{t+1}(\mu) = \beta_t(\mu \mid a_t, r_t)$.

Other bandit problems

Adversarial bandits

- ▶ Rewards are arbitrary.
- ▶ Compare with best arm in hindsight.

Continuous bandits

- ▶ Actions $a_t \in \mathbb{R}^d$
- ▶ Example: Lipschitz bandits where $|\rho(a) - \rho(a')| \leq \|a - a'\|$.

Contextual bandits (in particular linear)

- ▶ Contexts $x_t \in \mathbb{R}^d$
- ▶ Unknown parameters $\theta_a \in \mathbb{R}^d$
- ▶ For the linear case $\rho(x, a) = x^\top \theta_a$.

The Markov decision process

Bandit problems are not dynamic. We can generalise reinforcement learning to dynamical systems through the MDP formalism:

- ▶ Action space A .
- ▶ State space S .
- ▶ Transition kernel $s_{t+1} = j \mid s_t = s, a_t = a \sim P_\mu(j \mid s, a)$.
- ▶ Reward $r_t = \rho(s_t, a_t)$ (can also be random).
- ▶ Utility

$$U_t = \sum_{k=t}^T r_k.$$

Value functions

The state value function

For any given MDP μ and policy π we define

$$V_{\mu,t}^{\pi}(s) \triangleq \mathbb{E}_{\mu,t}^{\pi} [U_t \mid s_t = s]$$

The state-action value function

$$Q_{\mu,t}^{\pi}(s, a) \triangleq \mathbb{E}_{\mu,t}^{\pi} [U_t \mid s_t = s, a_t = a]$$

The optimal value functions

For an optimal policy π^*

$$V_{\mu,t}^* (s) \triangleq V_{\mu,t}^{\pi^*} (s) \geq V_{\mu,t}^{\pi} (s), \quad Q_{\mu,t}^* (s, a) \triangleq Q_{\mu,t}^{\pi^*} (s, a) \geq V_{\mu,t}^{\pi} (s, a)$$

The Bellman equations

State value function

$$\begin{aligned} V_{\mu,t}^{\pi}(s) &\triangleq \mathbb{E}_{\mu,t}^{\pi}[U_t \mid s_t = s] \\ &= \mathbb{E}_{\mu,t}^{\pi}[r_t + U_{t+1} \mid s_t = s] \\ &= \mathbb{E}_{\mu}^{\pi}[r_t \mid s_t = s] + \mathbb{E}_{\mu}^{\pi}[U_{t+1} \mid s_t = s] \\ &= \mathbb{E}_{\mu}^{\pi}[r_t \mid s_t = s] + \sum_{j \in S} \mathbb{E}_{\mu}^{\pi}[U_{t+1} \mid s_{t+1} = j] \mathbb{P}_{\mu}^{\pi}(s_{t+1} = j \mid s_t = s) \\ &= \mathbb{E}_{\mu}^{\pi}[r_t \mid s_t = s] + \sum_{j \in S} V_{\mu,t+1}^{\pi}(j) \mathbb{P}_{\mu}^{\pi}(s_{t+1} = j \mid s_t = s) \\ &= \mathbb{E}_{\mu}^{\pi}[r_t \mid s_t = s] + \sum_{j \in S} V_{\mu,t+1}^{\pi}(j) \sum_{a \in A} P_{\mu}(j \mid s, a) \pi(a \mid s_t). \end{aligned}$$

State-action value function

$$Q_{\mu,t}^{\pi}(s) = [\rho(s, a) + \sum_{j \in S} V_{\mu,t+1}^{\pi}(j) P_{\mu}(j \mid s, a)]$$

Optimal policies

Bellman optimality condition

The value function of the optimal policy satisfies this:

$$V_{\mu,t}^*(s) = \max_a [\rho(s, a) + \sum_{j \in S} V_{\mu,t+1}^*(j) P_{\mu}(j | s, a)]$$

Dynamic programming

To find V^* , Q^* , first initialise $V_{\mu,T}^*(s) = \max_a \rho(s, a)$. Then for $t = T - 1, T - 2, \dots, 1$:

$$Q_{\mu,t}^*(s, a) = \rho(s, a) + \sum_{j \in S} V_{\mu,t+1}^*(j) P_{\mu}(j | s, a).$$

$$V_{\mu,t}^*(s) = \max_a Q_{\mu,t}^*(s, a).$$

The optimal policy

The optimal policy is deterministic with:

$$a_t = \arg \max_a Q^*(s_t, a)$$

The Reinforcement Learning Problem

- ▶ Observe x_t
- ▶ Take action a_t
- ▶ Obtain reward r_t

Requirement for learning

- ▶ The model is not known
- ▶ Our policies must be **adaptive**

Reinforcement learning settings

Fully observable, discrete Markov problems

- ▶ $x_t = s_t$, a Markovian state, S, A finite.
- ▶ Optimal policies are Markov
- ▶ Can be solved efficiently with classical RL algorithms

Continuous Markov problems

- ▶ Requires function approximation
- ▶ Even when the model is known, hard to compute

Partially observable problems

- ▶ Sufficient statistics are not finite

Further resources

- ▶ The Sutton/Barto RL intro book
<http://incompleteideas.net/book/the-book-2nd.html>
- ▶ The Lattimore/Szepesvari bandit book
<https://tor-lattimore.com/downloads/book/book.pdf>
- ▶ The Dimitrakakis/Ortner RL book https://ilias.unibe.ch/goto_ilias3_unibe_file_2946650_download.html
- ▶ Reinforcement Learning Course at Neuchatel
<https://mcs.unibnf.ch/courses/reinforcement-learning-and-decision-making-under-uncertainty>
- ▶ OpenAI Gym <https://github.com/openai/gym/>